Ajinkya Nitin Phadtare

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Education

VESIT, Mumbai Dec 2022 - Present

B.E. in Electronics Engineering ***SGPA: 9.41/10 CGPA: 8.78/10***

Engineering Mathematics, Digital Logic Design, Microprocessors, Analog and Digital Circuits, Control Systems, Embedded Systems, Wireless Communication, VLSI and IoT

VJTI, Mumbai July 2019 - August 2022

Diploma in Electrical Engineering ***CGPA: 8.4/10***

Electrical Circuits, Power Systems, Electrical Machines, Control Systems, and Power Electronics.

Skills

Languages: C/C++, Python, Java, SQL, HTML, CSS

Technologies & Tools: Machine learning, Visual Studio, Flutter, PCB design, Embedded C programming, Visual Studio, EDA (Cadence), MATLAB/Simulink, Altium Designer, LT Spice, Arduino, Raspberry Pi, Keil, Eagle, KiCad, Thonny, I2C, SPI, UART, Wi-Fi, Bluetooth, IoT, VLSI, Multisim, Proteus

Work Experience

Panache Digilife Pvt Ltd, Mumbai November 2023 – July 2024

Product Engineer | Intern

- Executed rigorous performance evaluations to enhance product functionality and efficiency
- Led iterative design processes, fostering continuous improvement and ensuring products consistently met or exceeded performance benchmarks.
- Spearheaded the entire product Design, driving success from conceptualization through Research and development.

Techasia Mechatronics Pvt Ltd, Mumbai May 2023 - July 2023

Embedded Engineer | Intern

- Developed and implemented embedded systems for RS485 weighing machine project to interface with RS485 communication protocol.
- Conducted hardware testing and debugging to ensure accurate weight measurements.
- Collaborated with cross-functional teams to integrate hardware and software components

Project Work

Currency Detector and Model Training (2024): Engineered a currency detection system using sensors and machine learning algorithms. Included stages for image acquisition, preprocessing, and classification to accurately identify and authenticate currency notes. The project involved data classification, cleaning, and model training using Google Colab.

Petroleum Display Project using RS485 (2024): Developed a display system for petroleum stations using the RS485 communication protocol. The project involved interfacing various sensors and display units to monitor and display fuel levels, and other relevant data in real-time. Enforced efficient data transmission and error-checking mechanisms to ensure accurate and reliable communication between devices.

Smart Meter - Realtime Data Analysis and Monitoring (2024): Developed a smart metering system for real-time analysis and monitoring of electrical consumption. Integrated sensors for accurate energy measurement, implemented data transmission protocols, and utilized software algorithms for consumption pattern analysis. Aimed at optimizing energy efficiency and providing users with insights into their electricity usage.

Image Processing and Base64 Conversion (2023): Conducted testing and development on the AMB82 Board for face detection and object detection capabilities. Additionally, using the AMB82 Cam, I implemented photo capture via IP, integrated image captioning algorithms, and utilized base64 encoding for image data transmission. Successfully achieved live video streaming using RTSP, leveraging the boards' capabilities to enhance real-time image and video processing functionalities.

PCB Design for RCWL Motion Sensor Light Control (2023): Designed a printed circuit board (PCB) incorporating an RCWL motion sensor to automate the switching of lights. The PCB integrates the RCWL sensor with the necessary circuitry to detect motion and control the lighting system accordingly. This project aimed to enhance convenience and energy efficiency by automating light activation based on detected motion, utilizing robust PCB design principles for reliability and functionality.